

Lecture#07

Discrete Distribution Function

The cumulative distribution function (*CDF*) or simply the distribution function $F(x)$ of a discrete random variable X with probability function $f(x)$ defined over all real numbers x is the cumulative probability up to and including the point x . Symbolically, it is defined as follows:

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t).$$

The function $F(x)$ is monotonic increasing, i.e. $F(a) \leq F(b)$, whenever $a \leq b$. And the limit of F to the left is 0 and to the right is 1:

$$\lim_{x \rightarrow -\infty} F(x) = 0 \text{ and } \lim_{x \rightarrow +\infty} F(x) = 1.$$

The value of $F(x)$ at any point must be a number in the interval $0 \leq F(x) \leq 1$, because $F(x)$ is the probability of the event $(X \leq x)$.

Example: Consider the following table shows the probability distribution of a random variable X

X:	0	1	2	3
f(x):	20/120	60/120	36/120	4/120

Here,

$$F(0) = P(X = 0) = f(0) = \frac{20}{120}$$

$$F(1) = P(X \leq 1) = f(0) + f(1) = F(0) + f(1) = \frac{80}{120}$$

$$F(2) = P(X \leq 2) = f(0) + f(1) + f(2) = F(1) + f(2) = \frac{116}{120}$$

$$F(3) = P(X \leq 3) = f(0) + f(1) + f(2) + f(3) = F(2) + f(3) = 1$$

A tabular presentation of $F(x)$ is as follows:

X	< 0	0	1	2	3	> 3
f(x)	0	20/120	60/120	36/120	4/120	0
F(x)	0	20/120	80/120	116/120	1	1

A formal way of presenting the distribution function $F(x)$ is as follows:

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{20}{120}, & 0 \leq x < 1 \\ \frac{80}{120}, & 1 \leq x < 2 \\ \frac{116}{120}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Example: A coin is tossed three times. If X is the random variable representing the number of heads obtained, find the probability distribution of X and hence obtain $F(x)$.

Solution:

Sample space, $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

Let X be the number of head observed.

So, from the sample space,

$$[X = 3, 2, 2, 1, 2, 1, 1, 0]$$

$$X = 0, 1, 2, 3$$

So,

$$P(X = 0) = P(TTT) = \frac{1}{8}$$

$$P(X = 1) = P(HTT, THT, TTH) = \frac{3}{8}$$

And so on.

So, the probability distribution is:

X :	0	1	2	3
$f(x)$:	1/8	3/8	3/8	1/8

Therefore,

$$F(0) = f(0) = \frac{1}{8},$$

$$F(1) = f(0) + f(1) = \frac{4}{8},$$

$$F(2) = f(0) + f(1) + f(2) = \frac{7}{8},$$

$$F(3) = f(0) + f(1) + f(2) + f(3) = 1.$$

Thus,

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{8}, & 0 \leq x < 1 \\ \frac{4}{8}, & 1 \leq x < 2 \\ \frac{7}{8}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Example: A six-sided fair is rolled once. Obtain the probability distribution of the number of points on it and hence the distribution function.

Solution: Let X be the number of points on the die so that $X = 1, 2, 3, 4, 5, 6$. If P assigns equal mass to each of points on the die, then clearly

$$f(x) = P(X = x) = \frac{1}{6}$$

Then

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < 1 \\ \frac{1}{6}, & 1 \leq x < 2 \\ \frac{2}{6}, & 2 \leq x < 3 \\ \frac{3}{6}, & 3 \leq x < 4 \\ \frac{4}{6}, & 4 \leq x < 5 \\ \frac{5}{6}, & 5 \leq x < 6 \\ 1, & x \geq 6 \end{cases}$$

Exercises.

1. A random variable X takes the values 1, 2, 3, 4. Its pmf is

$$f(x) = kx; \quad x = 1, 2, 3, 4.$$

Calculate cdf or find the distribution function of X .

2. A random variable X has pmf:

$$P(X = x) = k(x^2 + 1); x = 0, 1, 2, 4.$$

Calculate cdf or find the distribution function of X .

3. A random variable X has pmf:

$$f(x) = k(x + 2); x = -1, 0, 1, 2, 4.$$

Calculate cdf or find the distribution function of X .

Continuous Probability Distribution

In dealing with continuous variable, $f(x)$ is usually called a probability density function (*pdf*) or simply density function. A formal definition of a probability density function may be presented as follows:

Definition: A probability density function is a non-negative function and is constructed so that the area under its curve bounded by the x -axis is equal to unity when computed over the range of x , for which $f(x)$ is defined.

The above definition leads to conclude that a *pdf* is one that possesses the following properties:

1. $f(x) \geq 0$
2. $\int_{-\infty}^{\infty} f(x)dx = 1$
3. $P(a < X < b) = \int_a^b f(x)dx$

Example: A random variable X has the following functional form:

$$f(x) = kx, \quad 0 < x < 4$$

$$= 0, \quad \text{elsewhere}$$

(i) Determine k for which $f(x)$ is a density function.

(ii) Find $P(1 < X < 2)$ and $P(X > 2)$.

Solution: (i) For $f(x)$ to be a density function, we must have $\int_{-\infty}^{\infty} f(x)dx = 1$.

Thus,

$$k \int_0^{\infty} xdx = 1$$

Or,

$$k \left[\frac{x^2}{2} \right]_0^4 = 1.$$

From which

$$k = \frac{1}{8}.$$

The complete density function is thus

$$f(x) = \frac{x}{8}, \quad 0 < x < 4$$

$$= 0, \quad \text{elsewhere}$$

(ii) Again,

$$P(1 < X < 2) = \frac{1}{8} \int_1^2 x dx = \left[\frac{x^2}{16} \right]_1^2 = \frac{3}{16}$$

and

$$P(X > 2) = \frac{1}{8} \int_2^4 x dx = \left[\frac{x^2}{16} \right]_2^4 = \frac{3}{4}.$$

Example: A continuous random variable X has the following density function:

$$f(x) = \frac{2}{27} (1 + x), \quad 2 < x < 5$$

$$= 0, \quad \text{elsewhere}$$

(i) Verify that it satisfies the condition $\int_{-\infty}^{\infty} f(x) dx = 1$,

(ii) Find $P(X < 4)$ and (iii) Find $P(3 < X < 4)$.

Solution: (i) Integrating between 2 and 5

$$\int_2^5 f(x) dx = \frac{2}{27} \int_2^5 (1 + x) dx = \frac{2}{27} \left[x + \frac{x^2}{2} \right]_2^5 = 1$$

showing that the given function is a density function.

(ii) Since the lower limit is 2, we integrate between 2 and 4 to evaluate $P(X < 4)$.

$$P(X < 4) = \frac{2}{27} \int_2^4 (1 + x) dx = \frac{2}{27} \left[x + \frac{x^2}{2} \right]_2^4 = \frac{16}{27}.$$

(iii) Evaluating the integral between 3 and 4, we obtain $P(3 < X < 4)$,

$$P(3 < X < 4) = \frac{2}{27} \int_3^4 (1 + x) dx = \frac{2}{27} \left[x + \frac{x^2}{2} \right]_3^4 = \frac{1}{3}.$$

Exercises.

1. A random variable X has the pdf:

$$f(x) = \begin{cases} k(4 - x^2); & 0 \leq x \leq 2 \\ 0 & ; \text{Otherwise} \end{cases}.$$

(i) Find k , (ii) Find $P(X = 1)$, (iii) $P(0.5 \leq X \leq 1.5)$ and Find (iv) $P(X \geq 1)$.

2. A random variable X has pdf:

$$f(x) = \begin{cases} k(x^2 + 1); & 0 \leq x \leq 3 \\ 0 & ; \text{Otherwise} \end{cases}.$$

(i) Find k , (ii) Find $P(X \geq 2)$, (iii) $P(1 \leq X \leq 1.5)$ and (iv) Find $P(X \leq 2)$.

3. A random variable X has pmf:

$$f(x) = kx^2 e^{-x}; x \geq 0.$$

(i) Find k , (ii) Find $P(X \leq 1)$, (iii) $P(1 \leq X \leq 3)$ and Find $P(X \geq 2)$.

(Note. $\int_0^\infty x^n e^{-x} dx = n!$)

Continuous Distribution Function

Exactly analogous to the distribution function of a discrete random variable, a continuous random variable as also a distribution function. This is defined as follows:

Definition: The cumulative distribution or distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is defined as

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

If the derivative of $F(X)$ exists, then

$$f(x) = \frac{d}{dx} F(x) = F'(x).$$

$F(x)$ has the following properties

1. $F(x) \geq 0$
2. $F(-\infty) = 0$
3. $F(+\infty) = 1$

$$\begin{aligned}
 4. P(a < X < b) &= P(a \leq X \leq b) = P(X \leq b) - P(X \leq a) = F(b) - F(a) \\
 &= \int_{-\infty}^b f(x)dx - \int_{-\infty}^a f(x)dx
 \end{aligned}$$

Example: If X has the density function

$$\begin{aligned}
 f(x) &= \frac{2}{27}(1+x), \quad 2 < x < 5 \\
 &= 0, \quad \text{elsewhere}
 \end{aligned}$$

Obtain the distribution function and hence find $F(3)$ and $F(4)$. Also verify $P(3 < x < 4) = F(4) - F(3)$.

Solution:

The distribution function is,

$$\begin{aligned}
 F(x) &= \frac{2}{27} \int_2^x (1+t)dt \\
 &= \frac{1}{27}(x^2 + 2x - 8) \\
 F(x) &= \begin{cases} 0, & x \leq 2 \\ \frac{1}{27}(x^2 + 2x - 8), & 2 < x < 5 \\ 1, & x \geq 5 \end{cases}
 \end{aligned}$$

For $x = 3$ and $x = 4$,

$$\begin{aligned}
 F(3) &= \frac{1}{27}(3^2 + 2 \times 3 - 8) = \frac{7}{27} \\
 F(4) &= \frac{1}{27}(4^2 + 2 \times 4 - 8) = \frac{16}{27}
 \end{aligned}$$

Hence,

$$F(4) - F(3) = \frac{16}{27} - \frac{7}{27} = \frac{9}{27} = \frac{1}{3}$$

Now, $P(3 < X < 4) = F(4) - F(3)$.

Example: Let X have the following probability density function:

$$f(x) = \begin{cases} x, & 0 < x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Obtain $F(x)$.

Solution:

$$F(x) = P(X \leq x)$$

$$= \begin{cases} 0, & x \leq 0 \\ \int_0^x t dt = \frac{x^2}{2}, & 0 < x \leq 1 \\ \int_0^1 t dt + \int_1^x (2-t) dt = 2x - \frac{x^2}{2} - 1, & 1 < x \leq 2 \\ 1, & x > 2 \end{cases}$$

Example: A box contains good and defective items. If an item drawn is good, the number 1 is assigned to the drawing; otherwise, the number 0 is assigned. Let p be the probability of drawing a good item at random.

Then

$$f(x) = P(X = x) = \begin{cases} 1 - p, & x = 0 \\ p, & x = 1 \end{cases}$$

Consequently, the distribution function of X is

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < 0 \\ 1 - p, & 0 \leq x < 1 \\ 1, & 1 \leq x \end{cases}$$

Exercises. Find the distribution function or cdf of the following pdfs.

1. A random variable X has the pdf:

$$f(x) = \begin{cases} k(4 - x^2); & 0 \leq x \leq 2 \\ 0 & ; \text{Otherwise} \end{cases}$$

2. A random variable X has pdf:

$$f(x) = \begin{cases} k(x^2 + 1); & 0 \leq x \leq 3 \\ 0 & ; \text{Otherwise} \end{cases}$$

3. A random variable X has pmf:

$$f(x) = \begin{cases} kx^2 e^{-x}; & 0 \leq x \leq 3 \\ 0 & ; \text{Otherwise} \end{cases}$$